Arenula

MCP server suite for the s&box game engine. Connects AI coding assistants to the editor in real time.

19

TOOLS

~120

ACTIONS

3

SERVERS

8-13K

TOKENS

ALL PHASES COMPLETE

Three Servers, One Suite

Editor

Scene manipulation, compilation, asset management, terrain editing. Runs inside the s&box editor as a C# plugin.

19 tools SSE on :8098 C#

API

Offline type and member reference. Indexes ~1,800 types and ~15,000 members from the s&box CDN schema.

8 tools stdio TypeScript

Docs

Narrative documentation fetcher. Scrapes docs.facepunch.com, converts to Markdown, caches with LRU.

3 tools stdio TypeScript

Editor Tools — 19 Omnibus Tools

Each tool uses an `action` parameter to select the operation. Grouped by resource, not operation type.

READ

scene

summary · hierarchy · statistics · find · find_in_radius · get_details · prefab_instances

7 actions

WRITE

gameobject

create · destroy · duplicate · reparent · rename · enable · set_tags · set_transform · batch_transform

9 actions

WRITE

component

add · remove · set_property · set_enabled · get_properties · get_types · copy

7 actions

NEW

compile

trigger · status · errors · generate_solution · wait

5 actions

WRITE

prefab

instantiate · get_structure · get_instances · break · update · create · save_overrides · revert · get_overrides

9 actions

READ

asset_query

browse search open get_dependencies get_model_info get_material_properties get_mesh_info get_bounds get_unsaved get_status get_json get_references

12 actions

NEW

asset_manage

create delete rename move save reload get_references

7 actions

WRITE

editor

select get_selected set_selected clear_selection frame_selection get_play_state start_play stop_play get_log save_scene save_scene_as undo redo console_list console_run open_code_file get_preferences set_preference

18 actions

WRITE

session

list set_active load_scene

3 actions

WRITE

lighting

create configure create_skybox set_skybox

4 actions

WRITE

physics

add_collider configure_collider add_rigidbody create_model_physics create_character_controller create_joint

6 actions

WRITE

audio

create configure

2 actions

WRITE

effects

create configure_particle configure_post_processing

3 actions

WRITE

camera

create configure

2 actions

WRITE

mesh

create_block create_clutter set_face_material set_texture_params set_vertex_position set_vertex_color set_vertex_blend get_info

8 actions

NEW

navmesh

create_agent create_area create_link generate get_status query_path

6 actions

ASYNC

cloud

search get_package mount

3 actions

WRITE

project

`get_collision` `set_collision_rule` `get_input` `get_info`

4 actions

NEW

terrain

`create` `configure` `get_info` `paint_material` `sync`

5 actions

Architecture

```
AI Client ──> HTTP/SSE ──> ArenulaMcpServer (inside s&box editor)
                                  |
                            RpcDispatcher
                                  |
                  ┌───────────────┼───────────────┐
                  |               |               |
          async tools      console path    main thread
         (compile, cloud)  (exception       (all other
          background        isolation)       scene tools)
                  |               |               |
              Handler         Handler         Handler
```

28 C# files: 8 Core infrastructure + 20 Handlers (one per tool). Async tools dispatch before `GameTask.MainThread()` to avoid deadlocks.

Arenula vs Ozmium (Original)

METRIC	OZMIUM	ARENULA
Tools	105	19
Total actions	105	~120
Token cost (schemas)	~42,000	~8-13,000
MCP best practices	None	Full Anthropic compliance
Code compilation	Not available	5 actions (trigger, status, errors, wait, generate_solution)
Asset CRUD	Read-only	Create, delete, rename, move, save
Prefab authoring	Instantiate only	Create, save overrides, revert, get overrides
Terrain	None	Create, configure, paint, sync
NavMesh	Place only	Generate, query paths, get status
Cloud assets	Deadlocked	Working (async dispatch)
Error messages	Generic strings	Structured with suggestions + fuzzy matching
Pagination	Truncation only	offset/limit with has_more metadata
Response format	Raw dumps	Canonical {id, name} tuples, concise/detailed modes
Cross-platform	Windows DllImport	Clean (no P/Invoke)

Design Principles

Following Anthropic's official MCP guidance.

Omnibus Tools

Related operations grouped with `action` enum. Fewer tools = less selection ambiguity for the LLM.

Rich Descriptions

3-4 sentences per tool. What it does, when to use it, when NOT to use it, what actions are available.

Trimmed Responses

Canonical {id, name} tuples. Pagination with offset/limit. 16K char truncation. Concise/detailed modes.

Actionable Errors

Structured {error, action, suggestion} with fuzzy type matching. "Did you mean 'Rigidbody'?"

Schema Safety

`additionalProperties: false` on every schema. Enum constraints on action parameters. Prevents hallucination.

Async Safety

Network tools (compile, cloud) dispatch before MainThread. Never `.GetAwaiter().GetResult()` on main thread.

Installation

1. Copy the Editor plugin

```
# Copy into your s&box project's Libraries folder Arenula-MCP/editor/ →
YourProject/Libraries/arenula_mcp/
```

2. Build the API and Docs servers

```
# API server cd Arenula-MCP/api && npm install && npm run build # Docs server cd
Arenula-MCP/docs && npm install && npm run build
```

3. Configure your .mcp.json

```
{ "mcpServers": { "editor": { "type": "sse", "url": "http://localhost:8098/sse" },
"api": { "command": "node", "args": ["path/to/Arenula-MCP/api/dist/index.js"] }, "docs":
{ "command": "node", "args": ["path/to/Arenula-MCP/docs/dist/index.js"] } } }
```

4. Open s&box

Arenula Editor compiles automatically and starts the MCP server on port 8098. Tools appear as `mcp__editor__*`, `mcp__api__*`, `mcp__docs__*`.

Based on Ozmium MCP Server by ozmium7 · API from sbox-api-mcp by sofianebel · Docs from sbox-docs-mcp

GPL-3.0 (Editor) · MIT (API, Docs)